

AI may have devastating impact on bees: study

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AI may have devastating impact on bees: study As artificial intelligence (AI) rapidly transforms industries and societies, a new report warns that this digital revolution carries serious, yet addressable, ecological risks. ‘The Digital Climate Footprint: From Global Data Centre Growth to Local Ecological Consequences’, produced by the Centre for AI, Trust and Governance at The University of Sydney, examines the impact of increased emissions from AI on Australian bees, and calls for urgent alignment between AI innovation and environmental stewardship. Introducing a ‘bee barometer’ framework, the report reveals how even small climate shifts can have major consequences on pollinators, food security and ecosystem integrity. The report estimates that AI-driven digital infrastructure, particularly energy-intensive model training and deployment processes, could contribute 4.8 to 15.4 gigatonnes of cumulative CO2 emissions globally between 2024 and 2035, driving global temperatures by 0.0026°C to 0.0084°C. “AI is reshaping how we work, live and govern, but its growth comes with an energy and environmental footprint that must be acknowledged,” said Dr Rob Nicholls from The University of Sydney. “We need a new model of AI innovation — one that is carbon-aware, ecologically informed, and responsible.” In Australia, bee-dependent agriculture, valued at \$4.6 billion annually, is already under pressure. Heat stress, bushfires and the spread of Varroa mites are putting both managed honeybees and native species such as stingless and carpenter bees at increasing risk. The increased emissions are driven by increased data centres as well as the broader AI ecosystem, including distributed computing, edge devices, manufacture and deployment of AI technologies, and the embodied energy of AI hardware, revealing environmental costs across the entire digital supply chain. To address these challenges, the report recommends: - conditional grid access for new hyperscale data centres (>100 MW); - adoption of carbon-aware computing and AI model efficiency standards; and - integrated partnerships across tech and conservation sectors. IBM Australia’s Executive Partner and Energy, Sustainability & Utilities Industry Lead APAC, Ravi Kumar Mandalika, said, “AI can be a powerful tool in the sustainability space, if we build it that way. “We’re seeing growing interest in AI systems that can help to improve efficiency, assess risk and enable smarter energy use. These tools, when designed responsibly, can help futureproof both business and the planet.” The report stresses that the years between 2024 and 2030 represent a critical window and that, with business leadership and foresight, Australia is well-positioned to become a global model for sustainable digital infrastructure and responsible AI innovation. “The future of AI and ecological health do not have to be in conflict,” Nicholls said. “This report provides a roadmap for building a digital future that serves people and the planet.” Image caption: Honeybee on a pea flower, Muogamarra Nature Reserve, Australia. Credit: iStock.com/RugliG Roadmap released for

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